INCH-POUND

MIL-P-45042F
27 November 1991
SUPERSEDING
MIL-P-45042E
21 November 1980

MILITARY SPECIFICATION

PUNCHING, SHEARING, NOTCHING, AND COPING MACHINE, METAL, MECHANICAL DRIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanically driven metalworking machines capable of performing punching, shearing, notching, and coping operations on steel bars, structural shapes, and plates of the dimensions and strengths specified herein.

1.2 Classification. The machine shall be of the following sizes. The size to be furnished shall be as specified (see 3.5 and 6.2). Size is determined by nominal punch capacity in tons and throat depth in inches.

Size	Nominal Capacity (Tons)	Throat Depth, Punch (Inches)
4	50	12
5	50	20
7a	70	15
7b	70	22
8a	90	19
9a	110	19
10a	135	29

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5213, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3445
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in Connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise Specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association Standards Department, Suite 201, 1500 King Street, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.5 - Iron Workers - Safety Requirements for the Construction, Care and Use. (DOD adopted)

ISO 54 Cylindrical Gears for General Engineering and for Navy Engineering - Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Battery march Park, Quincy, MA 02269 -9101.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspectiom (see 6.4) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operatio\ns in accordance with the requirements herein. Unless otherwise specified (see 6.2), the machine shall be provided with not less than four separate and permanent machining stations. Unless otherwise specified (see 6.2), the machine shall accomplish not less than the five operations of punching, notching, coping, flat bar shearing, and structural shape shearing. All stations shall be powered by the same drive. The franc and base of the machine shall be an integral unit designed to rest on a flat surface. Provisions suitable for hoisting the machine shall be provided. The base shall provide a means for securely bolting the machine to the floor or foundation. The design shall provide convenient, efficient and safe operation. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the sane system.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopadize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All operating controls for each work station shall be located and convenient to the operator. Controls shall include means for starting and stopping the machine and other necessary controlling devices. All operations of the machine shall be controlled by one or more anti-repeat

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

foot switches, except for punch spotting which may be by hand lever. When more than one anti-repeat foot switch Is required, the number, location, and operations control led by each foot switch shall be specified (see 6.2). These anti-repeat foot switches shall control performance of different operations at the same time and independent of each other.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910 and ANSI B11.5. If a conflict arises between OSHA 29 CFR 1910 and ANSI B11.5, OSHA 29 CER 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Safeguarding. Covers, guards, devices, or othermeans of safeguarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market shall be provided for the point-of-operation and all other parts of the machine that present safety hazards. If point-of-operation safeguarding is influenced by workpiece size and configuration, the procuring activity shall provide part design to the supplier of the machine (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the machine. All guards shall allow access to the guarded parts.

3.2.5.2 Noise level. Unless otherwise specified (see 6.2), noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response.

3.2.6 Mercury restriction, The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During the manufacture, service, transportation, or storage of the machine and its components the machine shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. The operation of the machine at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator’s health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.9 Lubrication. Unless otherwise specified (see 6.2), the machine shall be equipped with a manual lubrication system When specified, an automatic lubrication system shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, filler caps, and drain plugs shall be accessible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

3.2.10 Interchangeability. All parts having the saw manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be pertinently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of the drive train shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2001-B88, and AGMA 2000-A88 for the inch-pound system of measurennt or ISO 54 for the metric system of measurment. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

3.4 Components. The machine shall consist of not less than a base and frame, drive, punch, station, notcher station, coper station, bar shear station, structural shape shear station, and electrical equipment. The component stations may accomplish more than one operation in the same station.

3.4.1 Base and frame. The base and frame shall support the machining stations, drive system, and all other components specified herein. The base and franc enclosure shall have adequate strength, rigidity, and shock resistance to withstand all stresses produced when operating the machine at its maximum capacity without permanent deformation or premature failure. The machine shall be provided with a lifting eye or other means at or near its balance point on the frame for lifting the machine. The eye or lift point shall be capable of supporting five tines the weight of the machine, and its location shall not cause distortion to the franc when the machine Is suspended.

3.4.2. Drive. The machine shall be provided with a mechanical drive. The drive shall include a drive rotor, transmission, clutch, flywheel, drive shaft, connecting rod or cam, slide and ram.

3.4.2.1. Drive motor. The drive motor shall conform to the requirements of 3.4.8.1 and table I.

3.4.2.2. Transmission. The transmission shall consist of a gear or a Y-belt system. The V-belt system shall be capable of transmitting the required power without belt slippage. V-belt system shall have provisions for adjusting belt tension.

3.4.2.3 Clutch. A clutch or similar controlling device for engaging and disengaging the machine's operating functions shall be provided.

3.4.2.4 Flywheel. The flywheel shall be a balanced assembly having adequate mass and rotational speed to provide supplementary power to the rotor drive while performing work specified in table I. A constant speed of + 10 percent shall be maintained thru both the no-load and load phases of the operating cycle.

3.4.2.5 Drive shaft. The drive shaft shall transmit power developed by the drive train. The drive shaft shall be of adequate size and material to prevent deflection when subjected to maximum torque. The drive shaft shall be mounted on replaceable anti-friction bearings or bronze bushings of a size, type and precision class suitable for the application.

3.4.2.6 Connecting rod or cam. The connecting rod or cam shall be arranged to convert rotary motion to reciprocating motion in driving the rams of the machine. The rod or cam shall be mounted on replaceable anti-friction bearings or bronze bushings of a size, type and precision class suitable for the application.

3.4.2.7 Slide and ram. The slide and ram shall provide reciprocating motion for the machining operations. The ram shall be of solid or box-type construction, with reciprocating action guided by a slide or gibbed-way arrangement. The guide or gib shall be adjustable and replaceable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and characteristics.

	Size						
	4	5	7a	7b	8a	9a	10a
Punch Capacity:							
Punch Ram Pressure (Tons)	50	50	70	70	90	110	135
Throat Depth (inches)	12	20	15	22	19	19	29
1/ Strokes Per minute	44	44	44	44	44	44	42
Stroke length (inches)	15/16	15/16	1-1/8	1-1/8	1-3/16	1-1/4	1-1/2
Hole diameter (inches) in	1-1/16	1-1/16	1-1/16	1-1/16	1-3/16	1-7/16	1-5/16
Material thickness (inches)	1/2	5/8	3/4	3/4	13/16	7/8	1
Notching capacity (triangular):							
In angle sizes (inches)	2-1/2x5/16	3x5/16	3x3/8	3x3/8	3x7/16	3x7/16	4x3/8
Plate thickness (inches)	5/16	5/16	3/8	3/8	7/16	7/16	7/16
Depth of cut (inches)	2-3/8	2-3/4	2-3/4	2-3/4	2-3/4	3-1/4	3-3/4
Coping capacity (square):							
In angle sizes (inches)	3x5/16	3x3/8	3x3/8	3x1/2	3-1/2x7/16	4x1/2	4x5/8
In plate thicknesses (inches)	7/16	3/8	1/2	1/2	1/2	1/2	5/8
Depth of cut (inches)	2-3/4	3	3	3	3-1/8	3-1/2	3-1/2
Width of cut (inches)	1-9/16	2	2	2	2	2-1/4	2-1/4
Bar Shearing Capacity:							
Thickness (inches)	1/2	5/8	5/8	3/4	11/16	7/8	3/4
Flat bar size (inches)	4-3/4	7	4-1/2	9-1/4	6	6	6
Length of knives (inches)	8	10	9	12	10	14	12
Structural shapes shearing capacity:							
Round bar, diameter (inches)	1-3/8	1-5/8	1-3/4	2	2	2-1/4	2-9/16
Square bar (inches)	1-1/4	1-1/2	1-9/16	1-3/4	1-3/4	2	2-3/16
Angle, square cut (inches)	4x4x1/2	4x4x1/2	5x5x1/2	6x6x1/2	6x6x1/2	6x6x1/2	6x6x3/4
Angle, 45 degree miter (inches)	2-1/2x2-1/2x5/16	3x3x5/16	3-1/2x3-1/2x3/8	4x4x1/2	4x4x1/2	4x4x1/2	4x4x1/2
Drive:							
Motor Horsepower	2	3	3	3	4	7-1/2	11

NOTE: Based on cross section measurement of 60,000 PSI tensile strength material. Requirements shall be not less than shown. 1/ Strokes per minute are based upon the stroke length specified and the machine being cycled without a workpiece.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

3.4.3 Punch station. The punch station shall consist of a punch holder, die holder, and stripper. The punch station shall be capable of punching holes of the dimensions specified in table I. The station shall have a throat depth in accordance with table I. Means shall be provided to lower the punch for positioning over the workpiece.

3.4.3.1 Punch holder. The machine shall be provided with a punch holder for holding the punch in punching position under the punch ram. The punch holders shall be constructed of alloy steel and securely hold the punch in position. The punch holders shall assure allgnment of punches and dies. The punch holders shall be designed to accommodate standard round and shaped hole punches.

3.4.3.2 Die holder. The machine shall be provided with a die holder for holding the die in position. The die holders shall be constructed of hardened steel and precision ground to securely hold the dies in such a manner to prevent the dies from turning. The die holders shall be designed to accommodate standard round and shaped hole dies.

3.4.3.3 Stripper. The machine shall be provided with a stripper for automatically releasing the workpiece from the punch when the punch is withdrawn from the die. The stripper shall permit changing the punch without removing the stripper from the machine or the stripper shall be swung aside for changing the punch.

3.4.3.4 Punches and dies. When required (see 6.2), punches and dies of sizes and shapes specified shall be furnished. The punches and dies shall be tool steel, hardened, and ground for accuracy. Each punch and die shall be Permanently marked for size and identification.

3.4.4 Notcher station. The notcher station shall be capable of notching triangular cuts in metal to met the requirements of table I. The notcher shall be equipped with an adjustable gage designed to govern the depth of the cut. A guard shall be provided for protecting the operator when the notcher is operating, but is not being used. The blades of the notcher shall be removeable and replaceable.

3.4.5 Coper station. The coper station shall be capable of coping square cuts in metal to meet the requirements of table 1. The coper shall be provided with a gage for governing the depth of cut and a protective guard as specified for the notcher in 3.4.4. The blades of the coper shall be removeable and replaceable.

3.4.6 Flat bar shear station. The bar shear station shall consist of an upper moveable blade and a lower stationary blade, and shall be capable of shearing flat bar stock of the dimensions specified in table I. The bar shear shall be capable of shearing the maximum width capacity specified in table I to a quality that the camber of the sheared edge of the “drop-off” cut shall be 1/32 inch or less per 6 inch of bar width. The shear blades shall be not less than the length specified and shall be mounted in such a manner that they can be removed and replaced without the use of special tools. The flat bar shear shall be equipped with a stripper or hold down which shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

adjustable to accommodate stock of any dimensions within the capacity of the machine. The hold down shall be no farther than 4 inches from the shearing plane.

3.4.7 Structural shape shear. The structural shape and bar stock shear shall be capable of shearing structural shapes and bar stock of the dimensions specified in table I. The arrangement of the blades shall be such that structural angles fed for mitering cuts shall remain in a horizontal plane. Structural angle shapes shall be sheared at 90 degrees and mitered at 15 degrees, 30 degrees and 45 degrees, both leg up and leg down. The quality of the 'drop off" structural angle cuts shall be such that the two legs of the angle are neither apart by more than 91 degrees, 30 minutes, nor closer together than 88 degrees, 30 minutes. The size and arrangement of the blades shall be in accordance with the manufacturer's standard practice. The blades shall be mounted in such a manner that they can be removed and replaced without the use of special tools. The structural shape shear shall be equipped with an adjustable work piece stripper or hold-down, capable of accommodating stock of any shape or dimension within the capacity of the machine.

3.4.7.1 Angle iron muter gage. An angle iron miter gage shall 1 be furnished on the structural shape shear to allow shearing of angle iron at 15 degree, 30 degree, 45 degree, and 90 degree angles without the use of lay-out marks. The miter gage shall be adjustable to accommodate angle iron sizes from one inch x one inch up to the angle iron sizes specified in table I. Art adjustable stop shall be provided for locating the work in the desired position.

3.4.8 Electrical requirements. The machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 3-phase, 60-Hertz(Hz), dual 230/460 volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired as specified (see 6.2).

3.4.8.1 Motors. Motors shall meet the requirements of 3.4.8. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permenantly lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The rotor enclosure shall meet requirements for a drip-proof enclosure. All rotors shall conform to the requirements of NEMA MG-1.

3.4.8.2 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.9 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

not greater than 1 hour. The meter shall 1 be sealed to prevent the entrance of dust and moisture and shall 1 be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I for the size specified. Machine size and capacity shall be not less than the stated requirements of the size ordered.

3.6 Phisical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.7 Performance. The machine shall be capable ‘ of performing metal fabricating operations of punching, coping, bar shearing, structural shape shearing, and notching to conform with the requirements of table I. The machine shall be capable of performing these operations on structural steel workplaces having a 60,000 PSI minimum tensile strength.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer’s commercial machine shall be furnished.

3.9 Optional and accessory equipment. Optional and accessory equipment shall be furnished as specified (see 6.2). The equipment provided shall be functional and compatible to the size machine specified in table X.

3.10 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English laguage. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant

3.11.1 Lubrication plate or chart. When specified (see 6.2), a lubrication plate or chart shall be attached to each machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below:

Points of lubricant application
Type of lubricant
Servicing Interval
SAE number or military or Federal specification for each lubricant (if applicable)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

3.11.2 Namplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below:

Nomanclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest rotor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.12 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of the contract and shall be printed in the English language.

3.13 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall met all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contract-m's overall Inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The Inspection requirements specified herein are classified as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.

4.4 First article inspection. When a first article Inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance Inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 degrees to 104 degrees Farenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. Each machine shall be operated at no load for not less than one hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Performance tests. The machine shall be tested for conformance with 3.7 by performing the following tests.

4.8.3.1. Punch test. The machine shall be set up for punching operatioms. While utilizing the maximum size punch and die set specified in table I, not less than ten holes shall be punched In maximum thickness steel plate of the size specified under punching capacity in table I.

4.8.3.2 Notching test. The machine shall be set up for notching operations. Not less than one each of the specified shapes of the maximum sizes specified under notching capacity in table I shall be notched.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

4.8.3.3 Coping test. The machine shall be set up for coping operations. Not less than one each of the specified shapes of the maximum sizes specified under coping capacity in table I shall be coped.

4.8.3.5 Structural shape shearing test. The machine shall be set up for structural shape shearing operations. Not less than one each of the specified shapes of the maximum sizes specified in table I shall be sheared. Angle iron shapes shall be sheared at 90 degrees and mitered at angles of 15 degrees, 30 degrees and 45 degrees both leg up and leg down. The quality of the 'drop-off" structural angle cuts shall conform with 3.4.7.

4.8.4 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.3 shall be as specified and fully described (see 6.2).

4.8.5 Noise level test. Noise level measurements shall be made to ensure conformance with 3.2.5.2. Unless otherwise specified (see 6.2), the measurements shall be made at the operator's work station and each corner of the machine. Noise emitted by the machine when operating under a no-load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response.

4.8.6 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.8. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.8.7 Electromagnetic interference control test. Equipment requiring electromagnetic Interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains Information of a general or explanatory nature that may be helpful, but is not mandatory.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

6.1 Intended use. The machine covered by this specification is intended for use in punching, notching, coping, and shearing bars and structural shapes of various netals. This machine is intended for use in manufacturer's plants and military and contractor's repair shops.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, If required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Number of machining stations, if different (see 3.2).

f. Specify machining operations, if different (see 3.2).

g" System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

h. Specify two anti-repeat foot switches, if required (see 3.2.4).

i. Additional safety and health requirements, if required (see 3.2.5).

j. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5.1).

k. Specify noise level, if different (see 3.2.5.2).

1. Specify specific environmental protection requirements exceeding established EPA and OSHA requirements, when required (see 3.2.8).

m. Lubrication system, if different (see 3.2.9).

n. Specify surface finish requirements, if different (see 3.3.3).

o. Painting, If different (see 3.3.5).

p. If electromagnetic interference control is required, specify the equipment and subsystems class and the emission and susceptibility required (see 3.3.8).

q. Specify sizes and shapes of punches and dies, when required (see 3.4.3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

r. Voltage requirements, if different (see 3.4.8).

s. Initial voltage, as specified (see 3.4.8).

t. Motors, if different (see 3.4.8.1).

u. Worklight, if required (see 3.4.8.2).

v. Hour meter, if required (see 3.4.9).

w. Size and capacity, if different (see 3.5).

x. Specify physical size limitation, if required (see 3.6).

Y. Performance features required in addition to or in lieu of, the stated requirements (see 3.7.1).

z. Specify and fully describe optional equipment, if required (see 3.9).

aa. Specify applicable repair parts, if required (see 3.10).

bb. Lubrication plate, if required (see 3.11.1).

cc. Lubrication plate information, if different (see 3.11.1).

dd. Namplate, if different (see 3.11.2).

ee. Inspection conditions, if different (see 4.3).

ff. First article inspection, if different (see 4.4).

gg. Quality conformance inspection, if different (4.5).

hh. Acceptance test, if different (see 4.6).

ii. Additional performance tests, if required (see 4.8.4).

jj. Specify location of noise level test, if different (see 4.8.5).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

Reference	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of Items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired o tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System? International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

MIL-P-45042E (Covers both Mechanical and Hydraulic Ironworkers)	MIL-P-45042F (Covers only Hydraulic Ironworkers)
Size 1	Deleted
Size 2	Deleted
Size 3	Deleted
Size 4	Size 4
Size 5	Size 5
Size 6	Deleted
Size 7	Deleted
	Size 7a (new)
	Size 7b (new)
Size 8	Deleted
	Size 8a (new)
Size 9	Deleted
	Size 9a (new)
	Size 9b (new)
Size 10	Deleted
	Size 10a (new)

6.8 Changes from previous issues. Marginal notations are not used in this revision to identity changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard 1 eve 1 (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

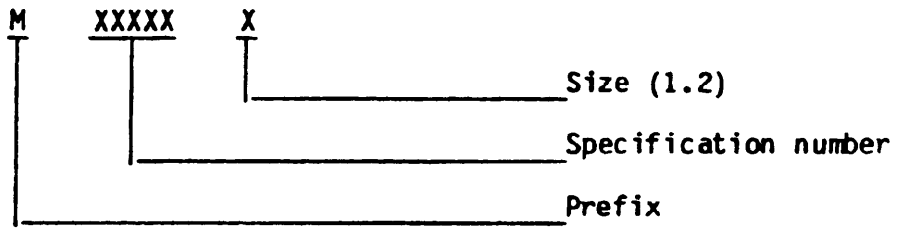
6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warrant Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. Where applicable, Arabic numbers shall be substituted for Roman numerals. Refer to the salient characteristics for origination of the correct letters or numbers to complete the PIN.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.



6.14 Subject term Kev word listing.

Combination metalworking machine
Ironworker (mechanical)
Metalworking machine
Steelworking machine

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Arm - AV
Navy - MC

Preparing activity:
DLA - IP

Project (3445-0116)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-45042F	2. DOCUMENT DATE (YYMMDD) 27 Nov 91

3. DOCUMENT TITLE
PUNCHING, SHEARING, NOTCHING, AND COPING MACHINE, METAL, MECHANICAL DRIVE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	b. TELEPHONE (Include Area Code) (1) Commercial: (901) 775-6269 (2) AUTOVON: 683-6269
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22102:152
Check the source to verify that this is the current version before use.